MEH-PPV blended CdS QDSSCs and Low Resistive Engineering of Nano CdS Films

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Abstract

Growing worldwide energy demand and concerns for costeffective technologies, have drawn the focus of the scientists and engineers towards hybrid solar cells which are synthesized by using the nanocomposites of inorganic semiconductors and conducting polymers.

Cadmium Sulphide (CdS) quantum dots (QDs) were deposited on nanostructured TiO_2 film usingSuccessive Ionic Layer Adsorption and Reaction (SILAR) method. The quantum dots were blended with a small quantity of MEH-PPV [Poly (2-methoxy-5-(2-ethylhexyloxy)-1, 4-phenylenevinylene] conductingpolymer powder mixed with polysulphide electrolyte. Nanostructured TiO_2 on FTO glassand Platinum on FTO are used as photoelectrode and Counter electrode respectively. High resolutionTransmission Electron Spectroscopy (HRTEM) image revealed that the CdS QDs adsorbedon TiO_2. The optical properties of CdS QD with MEH-PPV on TiO_2 film were studied by UV-Visabsorbance spectroscopy. The photovoltaic characteristics and impedance spectroscopic propertiesof CdS QDSSC were analyzed under air mass 1.5 illuminations.

Two basic requirements for the window material include: low electrical resistivity and high optical transmittance. Even though resistivity and optical transmittance strongly depend on the preparation conditions, undoped CdS films generally show high electrical resistivities. Thus, it is difficult to produce undoped CdS films with both low resistivity and high transmittance just by controlling preparation conditions. An effective way is to dope with suitable dopant and /or anneal these films. In the present work, chlorine has been chosen as dopant for CdS films by way of adding $CdCl_2$. Cu doping has also resulted in obtaining low resistive films with high transparency. A vivid description of the various techniques, such as thermal evaporation, chemical bath deposition (CBD), spray pyrolysis etc., will be presented in the keynote talk.

Biography

Dr. Amanullah Fatehmulla is Professor of Physics in the Department of Physics and Astronomy at King Saud University, Kingdom of Saudi Arabia. He is a member of the renewable energy and environment research group (Thin Films and Solar Cells). He is a member of department council as well as member of committee on Post Graduate Studies-Accreditation committee (Under Higher Studies & Research Committee). He received his PhD from Osmania University, Hyderabad, India in 1992. His research interests include thin film technology, solar cells, TCO (gas sensors), metal induced

crystallization (MIC) of amorphous silicon, renewable energy and environment, micro and nanostructured materials and nanocomposites. He has published over 60 research papers in International journals and conference proceedings. He has presented more than 30 research papers in International conferences and gave 8 invited talks. He has participated in several national and International conferences and workshops. His recent presentation/talk was in MRS- 60th Electronic Materials Conference held during June 27-29, 2018 at UCSB, Santa Barbara, USA and keynote lecturein 4thInternational Conference NANOCON 018 held during October 25-27, 2018 organized by BVU (Deemed University) in association with three Universities from USA in Pune, India. He was a visiting faculty during summer 2001 & 2002 (two months each) and worked in the photovoltaic division of the HiDEC/Electrical engineering department, University of Arkansas, Fayetteville, AR 72701, USA. His recent faculty visits include: visit to NIMS, Tsukuba, in Japan. He has been Principal Investigator and Co-Investigator for different scientific research projects. He is currently Co-Investigator for two research projects. As a teaching faculty, he has taught/teaching various undergraduate and postgraduate courses. He has supervised/supervising several graduate and undergraduate students. He is reviewer of several ISI journals and has been reviewing PhD and Master Thesis including scientific project proposals. He has been executing community services as selection committee member/ subject expert for post graduate teachers. His biographical record appeared for four consecutive times in "Who's Who in the World" starting from 2011. He is a member of many scientific bodies.

Photo



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